

SQL Theoretical Interview Questions & Answers for Data Science and Machine Learning

1. What is SQL?

SQL (Structured Query Language) is used to create, read, update, and delete data in relational databases.

2. Why is SQL important for Data Scientists?

SQL is essential for data extraction, cleaning, aggregation, feature engineering, and preparing datasets for ML models.

3. What is a relational database?

A database that stores data in tables with rows and columns, connected through keys.

4. What is a primary key?

A column or combination of columns that uniquely identifies each record in a table.

5. What is a foreign key?

A column that creates a relationship by referencing the primary key of another table.

6. What is normalization?

Normalization is the process of organizing data to reduce redundancy and improve integrity.

7. What are constraints in SQL?

Rules applied to maintain data accuracy such as NOT NULL, UNIQUE, CHECK, and DEFAULT.

8. What is NULL?

NULL represents missing or unknown data in a database.

9. Difference between WHERE and HAVING?

WHERE filters rows before aggregation, HAVING filters data after aggregation.

10. What is a JOIN?

JOIN combines rows from two or more tables based on a related column.

11. What are window functions?

Functions that perform calculations across related rows without collapsing results.

12. What is GROUP BY?

GROUP BY groups rows with similar values to apply aggregate functions.

13. What is a subquery?

A query nested inside another query to fetch intermediate results.

14. What is a CTE?

A Common Table Expression is a temporary named query used for clarity.

15. What is indexing?

Indexing improves query performance by enabling fast data retrieval.

16. OLTP vs OLAP?

OLTP handles real-time transactions; OLAP supports analytics and reporting.

17. What is data leakage?

Using future information while training ML models leading to inaccurate performance.

18. How SQL helps in feature engineering?

By generating aggregates, rolling metrics, and derived features.

19. What is denormalization?

Adding redundancy to improve read performance.

20. Difference between DELETE and TRUNCATE?

DELETE removes rows conditionally, TRUNCATE removes all rows instantly.